

Okore Joel Chidike

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Moscow, Russia

RESEARCH INTERESTS

Machine Learning, Cybersecurity, Graph Representation Learning, Graph Neural Networks, Temporal and Random Graphs, Network Anomaly Detection, Explainable AI, Distributed Systems

EDUCATION

- **Moscow Institute of Physics and Technology (MIPT)** Aug 2025 – Present (Expected Aug 2027)
MSc in Advanced Combinatorics Moscow, Russia
 - CGPA: 4.83/5.0
 - Relevant Coursework: Probabilistic Methods, Linear Algebra, Advanced Graph Theory, Discrete Geometry, Random Graphs
- **Innopolis University** Aug 2024 – Present (Expected Aug 2026)
MSc in Security and Network Engineering Innopolis, Russia
 - CGPA: 4.95/5.0
 - MSc Thesis: Real-Time Detection and Automated Response for Malicious HTTPS Traffic Using Explainable ML and SOAR Integration
- **Kazan National Research Technological University (KNRTU)** Sep 2020 – Aug 2024
BSc in Computer Science and Engineering Kazan, Russia
 - CGPA: 4.67/5.0
 - Scholarships: Bilateral Educational Agreement Scholarship (Russia–Nigeria), Ministry of Science and Higher Education of the Russian Federation - Tuition Fee Exemption

PUBLICATIONS & RESEARCH PAPERS

- **Explainable Machine Learning for Effective Malware Detection in Encrypted Network Traffic** [C.1]
J. C. Okore, I. Womoakor, and I. V. Kotenko. 2026 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology (USBEREIT)
 - Proposed a two-tier explainability pipeline for metadata-only encrypted-malware detection: a deterministic always-on tier (EBM / XGBoost contributions) for SOC triage and an on-demand tier (SHAP / LIME) for forensic investigation
 - Achieved $F1 \geq 0.9989$ across Random Forest, XGBoost, and a glass-box EBM, with EBM local explanations about $3.7\times$ faster than SHAP (3.5 ms median)
 - Showed that SHAP-guided reduction to a 9-feature consensus subset preserves near-perfect accuracy while improving explanation stability

EXPERIENCE

- **Innopolis University, Laboratory of Information Security** Jan 2026 - Present
Junior Researcher / Teaching Assistant Innopolis, Russia
 - Conduct practical and laboratory sessions for undergraduate and MSc courses in cybersecurity, network engineering, secure software development, digital forensics, incident response, and offensive security
 - Mentor students on network and systems security topics, including TCP/UDP programming, secure network design, vulnerability analysis, incident investigation, and security testing
 - Develop and maintain laboratory materials and reproducible environments for hands-on exercises in networking, cybersecurity, and secure systems
- **Innopolis University** May 2025 - Sep 2025
DevOps Engineer Intern Innopolis, Russia
 - Developed GitOps-based CI/CD workflows for Kubernetes environments using ArgoCD and Kustomize
 - Integrated automated security validation, monitoring, and observability pipelines using Prometheus, Grafana, SAST, and DAST tooling
 - Implemented automatic scaling of Jenkins agents on Kubernetes, and set up real-time monitoring using Prometheus/Grafana to track agent performance and cluster health.
- **CFSS Cyber Forensic Security Solutions** Feb 2024 - May 2024
Penetration Tester Intern Remote
 - Performed analysis of network and web application security mechanisms using tools such as Burp Suite, ZAP, and Postman to exploit logic vulnerabilities and API misconfigurations
 - Achieved privilege escalation in a Linux environment by exploiting incorrect settings, kernel vulnerabilities, and session hijacking techniques
 - Documented results and methodologies for technical and non-technical stakeholders and provided practical recommendations for remediating vulnerabilities, reducing the time to remediate vulnerabilities by 25%

• Laboratory of the Faculty of Control and Automation, KNRTU

Machine Learning Research Intern

Jan 2024 - Apr 2024

Kazan, Russia

- Developed machine learning models for anomaly detection on the KDDCup'99 dataset for binary and multiclass cyberattack classification
- Designed and optimized ML workflows using feature engineering, hyperparameter tuning, and comparative evaluation against IDS systems including Zeek, Snort, and Suricata
- Built an interactive framework for real-time traffic analysis and automated ML lifecycle management using ZenML

PROJECTS

• Aegis: Explainable ML Detection with Automated, Trust-Aware SOAR Response

2025 - 2026

MSc Thesis / Systems Research

NFStream, XGBoost, SHAP, Kafka, TheHive, Cortex, Wazuh

- Extended the XAI framework from the USBEREIT paper into a live system: retrained on NFStream flow features from 11,800 sessions across 25 real Windows malware families (F1 $\approx$ 0.97; 0.5% false-positive rate on out-of-distribution traffic)
- Turned the paper's qualitative feature-to-semantics annotation into an empirically validated feature-to-MITRE ATT&CK mapping, confirmed by four independent methods with 90–100% bootstrap stability and 87.5% end-to-end technique-assignment accuracy
- Co-developed the SOAR response layer: Cortex/MISP threat-intel enrichment, TheHive case automation, and a trust-gated decision matrix Wazuh Active-Response enforcement

• Automated Memory Forensics Pipeline for Real-Time Malware Detection and Response

2025

Security Automation

Wazuh, Volatility3, ML

- Developed an automated memory forensics framework integrating Wazuh, WinPMEM, and Volatility3 for malware detection and incidence response
- Designed ML-based classification workflows for malicious process identification using the CIC-MalMem2022 dataset
- Implemented automated feature extraction and SOAR-oriented response scripts for real-time analysis
- Simulated ransomware behavior in controlled environments for evaluation of detection and response workflows

• Real-Time ML-based Fraud Detection with Kafka Streams

2024

Streaming Analytics and Anomaly Detection

Kafka, Python, ML

- Developed a real-time transaction analysis pipeline using Apache Kafka and machine learning models for anomaly detection
- Implemented streaming data workflows for low-latency prediction and visualization
- Applied Random Forest-based classification for suspicious transaction detection in real-time environments
- Designed an end-to-end pipeline integrating data generation, streaming, inference, and monitoring

SKILLS

- **Programming Languages:** Python, Bash, C/C++, JavaScript, SQL
- **Machine Learning & Data Science:** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, SHAP, LIME
- **Cybersecurity & Systems:** Network Security, Malware Analysis, Intrusion Detection, Reverse Engineering, Distributed Systems, Linux Systems Programming
- **Infrastructure & Observability:** Docker, Kubernetes, Kafka, Git, CI/CD, Prometheus, Grafana, ELK Stack
- **Research Areas & Methods:** Machine Learning, Explainable AI, Graph-based Learning, Network Anomaly Detection, Statistical Learning, Real-Time Data Analytics

HONORS & AWARDS

• Open Doors Russian Scholarship Project

2023, 2024

Two-Time Winner

Russian Federation

- Awarded competitive international scholarship recognition for graduate study in Russia

• Innopolis University

2024 - 2026

Full Academic Scholarship

Innopolis, Russia

- Awarded full tuition scholarship for MSc studies in Security and Network Engineering

• Ministry of Science and Higher Education of the Russian Federation

2019 - 2024

Tuition Fee Exemption Scholarship

Russian Federation

- Awarded government-supported tuition exemption during undergraduate studies

• Bilateral Educational Agreement Scholarship

2019 - 2024

Scholarship Recipient

Russia - Nigeria

- Awarded international educational scholarship under the bilateral agreement between Russia and Nigeria

LANGUAGES

- English (Native/Fluent)
- Russian (B2 - Intermediate/Advanced)